

JIT, the resources are fetched just when they are needed, which means the applications get compiled when they are launched and are loaded into RAM. But compiling the entire code when we launch an application takes a lot of time, which translates into what we call lag. However, in this case, the entire code is not compiled on every application launch; rather, the part of the code that is needed to run the application is compiled every time and gets stored in a cache called the Dalvik-cache so that it can be reused. This cache gets optimised with every compilation overtime, and creates something like a tree of dependencies on every device.

Advanced Android users should know that they have to wipe the Dalvik-cache before they install a new ROM in their Android, because this tree of dependency now has to be reconstructed for the new system in the ROM.

But if the compiled application that is loaded into RAM is manually killed, the whole process of compilation has to be done again. Over time, the Dalvik-cache gets bigger and does not get cleaned; this takes up a lot of storage, which slows down the device.

How ART solves this

ART uses an AOT (Ahead-Of-Time) compiler. When an application is installed, the AOT compiler translates the entire code into machine code via compilation. So

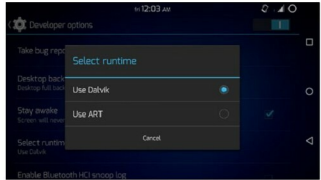


Figure 1: Enable ART



Figure 2: Keep tapping build number



Figure 3: Select runtime

this means the application doesn't need to be compiled repeatedly. This makes the process of launching and using the application faster and smoother, because the precompiled code (machine code) just needs to be executed and all the resources are readily available. Reducing the number of compilations also improves the battery life of the device. But compiling the entire code means installing the application will take more time and so will storage.

How to enable ART in KitKat devices

Go to **Settings > About phone**. Keep tapping 'Build number' until a toast message pops up.

Now go back to **Settings** and a new entry 'Developer options' can be found; turn it 'on', choose 'Select runtime' and then select ART.

The device needs to reboot, and might take a couple of minutes based on the number of applications you have installed because they need to get optimised for ART. 

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